Rigidity and finite-window failure for synchronized rounding defects

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Let

$$2^{a_j}x_{j+1} = 3x_j + 1, \qquad 2^{a_j}e_{j+1} = 3e_j - 1, \qquad a_j \in \mathbb{N}, \tag{1}$$

where every x_j is a positive integer and every e_j is a real number greater than 1. Put

$$q_j := \frac{3^j}{2^{A_j}}, \qquad A_j := \sum_{i<j} a_i, \qquad m_j := \lceil e_j \rceil, \qquad \theta_j := m_j - e_j, \tag{2}$$

and define

$$N_j := x_j + m_j, \qquad d_j := 2^{a_j}m_{j+1} - 3m_j + 1. \tag{3}$$

Then

$$d_j = 2^{a_j}\theta_{j+1} - 3\theta_j, \qquad -2 \le d_j \le 2^{a_j} - 1. \tag{4}$$

In the normalized divergent-orbit application we also have

$$q_j \longrightarrow \infty, \qquad \sum_{j=0}^{\infty} \frac{1}{x_j} < \infty, \qquad \sum_{j=0}^{\infty} \frac{1}{e_j} = \infty, \qquad \sum_{\substack{j\ge 0 \\ d_j \le 0}} \frac{1}{m_j} = \infty. \tag{5}$$

Lemma 1 (block defect cocycle)**.** *Fix $s \ge 0$ and $r \ge 1$. Write*

$$B_0 := 0, \qquad B_i := \sum_{h=0}^{i-1} a_{s+h}, \qquad M := 2^{B_r}, \qquad P := 3^r, \tag{6}$$

and put

$$C := \sum_{i=0}^{r-1} 3^{r-1-i}2^{B_i}, \qquad R := -\sum_{i=0}^{r-1} 3^{r-1-i}2^{B_i}d_{s+i}. \tag{7}$$

Then

$$\begin{aligned} Mx_{s+r} &= Px_s + C, \\ Mm_{s+r} &= Pm_s - C - R, \\ MN_{s+r} &= PN_s - R, \\ M\theta_{s+r} &= P\theta_s - R. \end{aligned} \tag{8}$$

If $d_{s+i} \le 0$ *for every* $0 \le i < r$*, then* $R \ge 0$ *and*

$$R \le P\theta_s < R + M. \tag{9}$$

Moreover, if $D := C + M$*, then*

$$D = Pm_s - \big(R + M(m_{s+r} - 1)\big), \tag{9a}$$

and, for every $0 \le t \le r$*,*

$$\boxed{3^t \mid D \quad \Longleftrightarrow \quad 3^t \mid R + M(m_{s+r} - 1).} \tag{9b}$$

The integer on the right also has the exact companion form

$$R + M(m_{s+r} - 1) = P\theta_s + M(e_{s+r} - 1) > 0. \tag{9c}$$

Proof. The first identity in (8) is the usual affine iteration of the first recurrence in (1). From (3) we have

$$2^{a_j} m_{j+1} = 3m_j - 1 + d_j.$$

Iterating this recurrence gives

$$Mm_{s+r} = Pm_s - C + \sum_{i=0}^{r-1} 3^{r-1-i} 2^{B_i} d_{s+i} = Pm_s - C - R.$$

Adding the first two identities proves the identity for N. Iteration of (4) proves the identity for θ. If every displayed defect is nonpositive, then $R \ge 0$. The inequalities $0 \le \theta_{s+r} < 1$ in the last identity of (8) give (9).

Rearranging the identity for m in (8) gives (9a). Since $P = 3^r$, reduction modulo 3^t proves (9b). Finally, the last identity in (8) and $m_{s+r} - 1 - \theta_{s+r} = e_{s+r} - 1$ give (9c). Its strict positivity follows from $e_{s+r} > 1$. □

For a nonzero integer z, we write $\nu_2(z) := v_2(|z|)$.

Theorem 2 (defect-switch valuation conservation)**.** *Assume (1)–(5), and put*

$$c_j := 1 - d_j, \qquad Z_j(c) := cx_j + m_j. \tag{10}$$

There is an index J such that $Z_j(c) \neq 0$ for every integer c and every $j \geq J$. For every $j \geq J$ we have

$$\boxed{2^{a_j} Z_{j+1}(c_j) = 3Z_j(c_j).} \tag{11}$$

Partition the tail into its maximal constant-defect runs. Let the k-th run begin at s_k, end at s_{k+1}, have color c_k, and have total exponent

$$B_k := \sum_{j=s_k}^{s_{k+1}-1} a_j. \tag{12}$$

Define

$$\begin{aligned} u_k &:= \nu_2\big(Z_{s_k}(c_k)\big), \\ v_k &:= \nu_2\big(Z_{s_{k+1}}(c_k)\big), \\ g_k &:= \nu_2\big(Z_{s_{k+1}}(c_{k+1})\big) - \nu_2\big(Z_{s_{k+1}}(c_k)\big). \end{aligned} \tag{13}$$

Then every run is finite,

$$\boxed{B_k = u_k - v_k,} \tag{14}$$

and, for every $K \geq 0$,

$$\boxed{\sum_{k=0}^{K} B_k = u_0 - v_K + \sum_{k=0}^{K-1} g_k.} \tag{15}$$

In particular,

$$\sum_{k=0}^{K-1} g_k \longrightarrow \infty. \tag{16}$$

At a switch, put $c := c_k$, $c' := c_{k+1}$, $x := x_{s_{k+1}}$, $m := m_{s_{k+1}}$, and $h := \nu_2(c' - c)$. If

$$v := \nu_2(cx + m),$$

then

$$g_k = \begin{cases} 0, & v < h, \\ h - v, & v > h, \\ \nu_2\left(\dfrac{cx+m}{2^h} + \dfrac{c'-c}{2^h}x\right), & v = h. \end{cases} \tag{17}$$

Consequently, a switch has positive gain if and only if

$$\nu_2(cx + m) = \nu_2(c' - c). \tag{18}$$

Proof. Put $X_j := x_j + 1$ and $E_j := e_j - 1$. If $\gamma_j := 2^{a_j} - 2$, then (1) gives

$$2^{a_j} X_{j+1} = 3X_j + \gamma_j, \qquad 2^{a_j} E_{j+1} = 3E_j - \gamma_j.$$

Therefore,

$$\frac{E_j}{X_j} - \frac{E_{j+1}}{X_{j+1}} = \frac{\gamma_j (E_j + X_j)}{X_j (3X_j + \gamma_j)} \geq 0.$$

The ratio E_j/X_j has a limit. If this limit were positive, then $e_j \gg x_j$ on a tail, which would make $\sum_j 1/e_j$ converge by (5). Thus $E_j/X_j \to 0$. Since $x_j \to \infty$ and $m_j < e_j + 1$, we obtain

$$\frac{m_j}{x_j} \longrightarrow 0. \tag{19}$$

Choose J so that $m_j < x_j$ for every $j \geq J$. If $c \geq 0$, then $Z_j(c) > 0$. If $c \leq -1$, then $Z_j(c) < 0$. Hence these linear forms do not vanish on the chosen tail.

By the definition of c_j, equation (3) becomes

$$2^{a_j} m_{j+1} = 3m_j - c_j.$$

Multiplying the positive recurrence in (1) by c_j and adding gives

$$\begin{aligned} 2^{a_j} Z_{j+1}(c_j) &= c_j(3x_j + 1) + 3m_j - c_j \\ &= 3Z_j(c_j), \end{aligned}$$

which proves (11).

Iteration through one constant-color run gives

$$2^{B_k} Z_{s_{k+1}}(c_k) = 3^{s_{k+1} - s_k} Z_{s_k}(c_k). \tag{20}$$

Taking 2-adic valuations proves (14). In particular, no constant-color tail is possible, since its accumulated exponent would eventually exceed the fixed valuation at its initial state. Thus every maximal run is finite.

By definition, $g_k = u_{k+1} - v_k$. Summing (14) and inserting these differences gives (15). Since every $B_k \geq 1$ and $v_K \geq 0$, equation (15) gives (16).

Finally,

$$c'x + m = (cx + m) + (c' - c)x. \tag{21}$$

The integer x is odd, so the second term on the right has valuation h. If $v \neq h$, the valuation of the sum is $\min\{v, h\}$. If $v = h$, division by 2^h leaves a sum of two odd integers. This proves (17) and (18). □

Corollary 3 (eventually constant defects are impossible)**.** *Assume (1)–(5). There do not exist $J \geq 0$ and $\delta \in \mathbf{Z}$ such that*

$$d_j = \delta \qquad (j \geq J). \tag{22}$$

In particular, none of the three nonpositive defects $-2, -1, 0$ can be eventually constant.

Proof. This is the constant-color tail excluded in the proof of the theorem. □

Corollary 4 (constant-defect terminal decoder)**.** *We use the convention $v_3(0) := \infty$. Let $[s, s+r)$ be a constant-defect run with*

$$d_j = 1 - c, \qquad c \in \{1, 2, 3\}, \qquad s \leq j < s + r.$$

Then

$$\boxed{3^r \mid cx_{s+r} + m_{s+r}.} \tag{D1}$$

If $c \in \{1, 2\}$, then

$$\boxed{\min\{v_3(x_{s+r} + 1), r\} = \min\{v_3(m_{s+r} - c), r\}.} \tag{D2}$$

For $c = 3$, we have

$$\boxed{\min\{v_3(x_{s+r} + 1) + 1, r\} = \min\{v_3(m_{s+r} - 3), r\}.} \tag{D3}$$

In particular, a constant $d = -1$ run ending at $m_{s+r} = 2$ satisfies $3^r \mid x_{s+r} + 1$.

Proof. Iteration of (11) through the run gives

$$2^{B_r}\big(cx_{s+r} + m_{s+r}\big) = 3^r(cx_s + m_s).$$

Since 2^{B_r} is coprime to 3, this proves (D1). Reducing (D1) after writing

$$cx_{s+r} + m_{s+r} = c(x_{s+r} + 1) + (m_{s+r} - c)$$

proves (D2). When $c = 3$, the same congruence compares $3(x_{s+r} + 1)$ with $m_{s+r} - 3$, which proves (D3). □

Theorem 5 (switch-carry terminal decoder)**.** *Fix a block of length $r \geq 1$. For $0 \leq i \leq r$, put*

$$x_i := x_{s+i}, \qquad m_i := m_{s+i}, \qquad c_i := 1 - d_{s+i},$$

and put

$$B_0 := 0, \qquad B_i := \sum_{h=0}^{i-1} a_{s+h}, \qquad M := 2^{B_r}.$$

Define the synchronized switch carry by

$$\mathscr{S}_{s,r} := \sum_{i=1}^{r} 3^{r-i} 2^{B_i} (c_i - c_{i-1}) x_i. \tag{S1}$$

Then

$$\boxed{M(c_r x_r + m_r) = 3^r (c_0 x_0 + m_0) + \mathscr{S}_{s,r}.} \tag{S2}$$

Consequently,

$$\boxed{M c_r (x_r + 1) \equiv \mathscr{S}_{s,r} - M(m_r - c_r) \pmod{3^r}.} \tag{S3}$$

If $c_r \neq 0$, then

$$\boxed{\min\{v_3(x_r + 1) + v_3(c_r), r\} = \min\{v_3(\mathscr{S}_{s,r} - M(m_r - c_r)), r\}.} \tag{S4}$$

Suppose that the block contains at least one color switch, and let

$$\ell := \max\{1 \leq i \leq r : c_i \neq c_{i-1}\}.$$

If

$$3 \nmid (c_\ell - c_{\ell-1}) x_\ell, \tag{S5}$$

then

$$\boxed{v_3(\mathscr{S}_{s,r}) = r - \ell.} \tag{S6}$$

Proof. Equation (11), followed by the change from color c_i to color c_{i+1}, gives

$$\begin{aligned} 2^{a_{s+i}} Z_{i+1}(c_{i+1}) &= 2^{a_{s+i}} Z_{i+1}(c_i) + 2^{a_{s+i}} (c_{i+1} - c_i) x_{i+1} \\ &= 3 Z_i(c_i) + 2^{a_{s+i}} (c_{i+1} - c_i) x_{i+1}. \end{aligned}$$

Iteration proves (S2). Writing

$$c_r x_r + m_r = c_r (x_r + 1) + (m_r - c_r)$$

and reducing (S2) modulo 3^r proves (S3). Since M is a unit modulo 3^r, comparison of truncated valuations proves (S4).

By the definition of the last switch,

$$\mathscr{S}_{s,r} = 3^{r-\ell}\left(2^{B_\ell}(c_\ell - c_{\ell-1})x_\ell + 3H\right)$$

for some integer H. Under (S5), the expression in parentheses is a unit modulo 3. This proves (S6). □

Corollary 6 (last-switch matched-shell law)**.** *Use the notation and hypotheses of the switch-carry terminal decoder. Assume that $c_r \neq 0$, that the block contains a color switch, and that its last switch ℓ satisfies (S5). Put*

$$t := r - \ell, \qquad h := v_3(m_r - c_r), \qquad \delta := \min\{v_3(x_r + 1) + v_3(c_r), r\}, \tag{L1}$$

where $v_3(0) := \infty$. If $h \neq t$, then

$$\boxed{\delta = \min\{h, t\}.} \tag{L2}$$

If $h = t$, then $\delta \geq t$, and for every $1 \leq q \leq r - t$ we have

$$\boxed{\delta \geq t + q \quad \Longleftrightarrow \quad \frac{\mathscr{S}_{s,r}}{3^t} \equiv \frac{M(m_r - c_r)}{3^t} \pmod{3^q}.} \tag{L3}$$

In particular,

$$\boxed{\delta > t \quad \Longleftrightarrow \quad \begin{cases} v_3(m_r - c_r) = t, \\ 2^{B_\ell}(c_\ell - c_{\ell-1})x_\ell \equiv \dfrac{M(m_r - c_r)}{3^t} \pmod 3. \end{cases}} \tag{L4}$$

If $3 \nmid c_r$, then $\delta = \min\{v_3(x_r + 1), r\}$, so (L2)–(L4) decode the terminal factor $x_r + 1$ itself.

Proof. By (S6),

$$v_3(\mathscr{S}_{s,r}) = t. \tag{L5}$$

Since M is a power of 2, we also have

$$v_3\big(M(m_r - c_r)\big) = h. \tag{L6}$$

If $h \neq t$, the two terms in the right side of (S3) have distinct 3-adic valuations. Hence

$$v_3\big(\mathscr{S}_{s,r} - M(m_r - c_r)\big) = \min\{h, t\} < r,$$

and (S4) proves (L2).

Now suppose that $h = t$. Both terms in the same difference are divisible by 3^t, so (S4) gives $\delta \geq t$. For $1 \leq q \leq r - t$, the same identity gives

$$\begin{aligned} \delta \geq t + q \quad &\iff \quad 3^{t+q} \mid \mathscr{S}_{s,r} - M(m_r - c_r) \\ &\iff \quad \frac{\mathscr{S}_{s,r}}{3^t} \equiv \frac{M(m_r - c_r)}{3^t} \pmod{3^q}. \end{aligned}$$

This proves (L3). Finally, the factorization used in the proof of (S6) gives

$$\frac{\mathscr{S}_{s,r}}{3^t} \equiv 2^{B_\ell}(c_\ell - c_{\ell-1})x_\ell \pmod 3.$$

Taking $q = 1$ in (L3), and combining the result with (L2), proves (L4). The last assertion follows directly from $3 \nmid c_r$. □

Corollary 7 (nonpositive-block reciprocal shell)**.** *Use the notation of the switch-carry terminal decoder, and suppose that*

$$c_i \in \{1, 2, 3\} \qquad (0 \leq i \leq r). \tag{N1}$$

If the block contains a color switch, then its last switch ℓ automatically satisfies (S5). Put

$$t := r - \ell, \qquad \delta := \min\{v_3(x_r + 1) + v_3(c_r), r\}. \tag{N2}$$

If $t \geq 1$ and $\delta > t$, then

$$\boxed{v_3(m_r - c_r) = t, \qquad m_r \geq 3^t + c_r, \qquad \frac{1}{m_r} < 3^{-t}.} \tag{N3}$$

Consequently, let a constant-color run inside the block begin at a switch $u \geq 1$ and end at $v \leq r$. For $u \leq j \leq v$, put

$$t_j := j - u, \qquad \delta_j := \min\{v_3(x_{s+j} + 1) + v_3(c_j), j\}.$$

Then

$$\boxed{\sum_{\substack{u \leq j \leq v \\ t_j \geq T,\ \delta_j > t_j}} \frac{1}{m_{s+j}} < \sum_{t=T}^{\infty} 3^{-t} = \frac{3^{1-T}}{2} \qquad (T \geq 1).} \tag{N4}$$

Proof. For every $j \geq 1$, the first recurrence in (1) gives

$$2^{a_{j-1}} x_j = 3x_{j-1} + 1 \equiv 1 \pmod 3.$$

Thus $3 \nmid x_j$. At a color switch inside a block satisfying (N1), the nonzero difference $c_i - c_{i-1}$ belongs to $\{-2, -1, 1, 2\}$. It is also a unit modulo 3. Since the global index $s + \ell$ is positive, (S5) follows.

If $\delta > t$, equation (L4) gives $v_3(m_r - c_r) = t$. When $t \geq 1$, the nonzero integer $m_r - c_r$ is a multiple of 3^t. Since $m_r \geq 2$ and $1 \leq c_r \leq 3$, it cannot be a negative multiple of 3^t. Hence $m_r - c_r \geq 3^t$, which proves (N3).

Apply (N3) to each block prefix ending at $s + j$. After the switch at u, the run age t_j takes each nonnegative integer at most once. Summing the resulting bounds over the selected endpoints and then enlarging to the full geometric tail proves (N4). □

Corollary 8 (global late-shell reduction)**.** *Assume (1)–(5). Partition the set*

$$\{j \geq 0 : d_j \leq 0\}$$

into its maximal consecutive blocks $I_b = [\alpha_b, \beta_b]$, allowing an infinite right endpoint. Partition each I_b into maximal constant-color runs

$$I_{b,k} = [u_{b,k}, v_{b,k}] \qquad (0 \leq k \leq K_b),$$

listed in increasing order, where $K_b \in \{0, 1, 2, \ldots\} \cup \{\infty\}$. For $k \geq 1$ and $j \in I_{b,k}$, put

$$t_j := j - u_{b,k}, \qquad \delta_{b,j} := \min\{v_3(x_j + 1) + v_3(c_j), j - \alpha_b\}. \tag{G1}$$

Choose integers $T_{b,k} \geq 1$ for $k \geq 1$ such that

$$\sum_b \sum_{k=1}^{K_b} 3^{-T_{b,k}} < \infty. \tag{G2}$$

Then

$$\boxed{\sum_b \sum_{k=1}^{K_b} \sum_{\substack{j \in I_{b,k} \\ t_j \geq T_{b,k},\ \delta_{b,j} > t_j}} \frac{1}{m_j} < \infty.} \tag{G3}$$

Consequently, at least one of the following three sets has divergent reciprocal mass:

$$\begin{aligned}\mathcal{E} &:= \bigcup_b I_{b,0},\\ \mathcal{S} &:= \bigcup_b \bigcup_{k=1}^{K_b} \{j \in I_{b,k} : t_j < T_{b,k}\},\\ \mathcal{L} &:= \bigcup_b \bigcup_{k=1}^{K_b} \{j \in I_{b,k} : t_j \geq T_{b,k},\ \delta_{b,j} \leq t_j\}.\end{aligned} \tag{G4}$$

That is,

$$\boxed{\sum_{j\in\mathcal{E}} \frac{1}{m_j} = \infty, \quad \text{or} \quad \sum_{j\in\mathcal{S}} \frac{1}{m_j} = \infty, \quad \text{or} \quad \sum_{j\in\mathcal{L}} \frac{1}{m_j} = \infty.} \tag{G5}$$

Proof. For $k \geq 1$, the run $I_{b,k}$ begins at a color switch contained in the nonpositive block I_b. Apply (N4) with $s = \alpha_b$, $u = u_{b,k} - \alpha_b$, $v = v_{b,k} - \alpha_b$, and $T = T_{b,k}$. We obtain

$$\sum_{\substack{j\in I_{b,k}\\ t_j\geq T_{b,k},\ \delta_{b,j}>t_j}} \frac{1}{m_j} < \frac{3^{1-T_{b,k}}}{2}.$$

Summing this inequality and using (G2) proves (G3).

The four sets consisting of the summation set in (G3) and the three sets in (G4) partition all nonpositive-defect indices. Their total reciprocal mass diverges by (5), whereas the first set has finite mass by (G3). Therefore the union of the three sets in (G4) has divergent mass. A finite union of convergent nonnegative series is convergent, so at least one of the three displayed series in (G5) diverges. □

Proposition 9 (unbounded entrance mass in finite windows)**.** *For every $k \geq 1$, let λ_k be the unique odd integer in $\{1, 3, 5, \ldots, 17\}$ such that*

$$\lambda_k 2^{k+4} \equiv 13 \pmod 9. \tag{E1}$$

Equivalently, according to the residue of k modulo 6, we have

$$\lambda_k = 7, 17, 13, 11, 1, 5. \tag{E2}$$

The strict-barrier word

$$w_k := (1, 2, 1^k) \tag{E3}$$

has a positive odd integer realization and a synchronized real companion such that

$$1 < e_j < 2, \qquad m_j = 2 \qquad (0 \le j \le k+2), \tag{E4}$$

and

$$(d_0, d_1, d_2, \ldots, d_{k+1}) = (-1, 3, -1^k). \tag{E5}$$

Thus the second maximal nonpositive-defect block is one entrance run of length k, and

$$\boxed{\sum_{j=2}^{k+1} \frac{1}{m_j} = \frac{k}{2}.} \tag{E6}$$

In particular, entrance-run reciprocal mass has no uniform finite-window bound, even for strict-barrier words with a positive ordinary realization. No one nonnegative ordinary seed realizes all of the nested words w_k.

Proof. Since 2 is a unit modulo 9, equation (E1) determines one residue class modulo 9. Exactly one of its representatives in $\{1, 2, \ldots, 17\}$ is odd. This proves existence and uniqueness, and direct reduction gives (E2).

Define

$$\begin{aligned} x_0 &:= \frac{\lambda_k 2^{k+4} - 13}{9}, \\ x_1 &:= \frac{\lambda_k 2^{k+3} - 5}{3}, \\ x_{2+i} &:= \lambda_k 2^{k+1-i} 3^i - 1 \qquad (0 \le i \le k). \end{aligned} \tag{E7}$$

Equation (E1) makes x_0 an integer. Reducing (E1) modulo 3 also shows that x_1 is an integer. All the integers in (E7) are positive and odd. Direct substitution gives

$$2x_1 = 3x_0 + 1, \qquad 4x_2 = 3x_1 + 1, \qquad 2x_{3+i} = 3x_{2+i} + 1 \quad (0 \le i < k). \tag{E8}$$

Since every state on the right side of a division in (E8) is odd, the displayed powers of 2 are exact. Hence (E7) realizes w_k.

For a prefix of length n, the accumulated exponent is 1 when $n = 1$ and $n + 1$ when $2 \le n \le k + 2$. We have $2 < 3$, and

$$2^{n+1} < 3^n \qquad (n \ge 2),$$

because the inequality holds at $n = 2$ and its left-to-right ratio is multiplied by $2/3$ when n increases. Thus every nonempty prefix is a strict barrier.

We now define the real companion by

$$\begin{aligned} e_{2+i} &:= 1 + \frac{1}{8}\left(\frac{2}{3}\right)^{k-i} \qquad (0 \le i \le k), \\ e_1 &:= \frac{5}{3} + \frac{1}{6}\left(\frac{2}{3}\right)^{k}, \\ e_0 &:= \frac{13}{9} + \frac{1}{9}\left(\frac{2}{3}\right)^{k}. \end{aligned} \tag{E9}$$

One can check directly that

$$2^{a_j} e_{j+1} = 3e_j - 1 \qquad (0 \le j \le k+1).$$

Moreover,

$$e_{2+i} \le \frac{9}{8}, \qquad e_1 \le \frac{11}{6}, \qquad e_0 \le \frac{14}{9},$$

and every number in (E9) is greater than 1. This proves (E4).

Since $m_j = 2$, equation (3) becomes

$$d_j = 2^{a_j+1} - 5.$$

The exponents in (E3) therefore give (E5), and (E6) follows. Finally, suppose that one nonnegative integer realized every w_k. After the first two steps, repeated use of the exponent-one recurrence would give

$$x_{2+i} + 1 = \frac{3^i}{2^i}(x_2 + 1) \qquad (i \ge 0).$$

Thus $2^i \mid x_2 + 1$ for every i, which forces $x_2 = -1$. This contradicts nonnegativity. □

Corollary 10 (run-mass valuation budget)**.** *Let $[s, u]$ be a constant-color run with*

$$c_j = c \in \{1, 2, 3\} \qquad (s \le j \le u).$$

Put

$$B := \sum_{j=s}^{u} a_j, \qquad U := \nu_2(cx_s + m_s), \qquad V := \nu_2(cx_{u+1} + m_{u+1}), \tag{V1}$$

and define

$$\kappa_1 := \kappa_3 := \frac{1}{3}, \qquad \kappa_2 := \frac{1}{2}. \tag{V2}$$

Then

$$\boxed{B = U - V, \qquad \sum_{j=s}^{u} \frac{1}{m_j} \le \kappa_c B = \kappa_c (U - V) \le \kappa_c U.} \tag{V3}$$

Suppose in addition that $s \ge 1$, $c_{s-1} \le 0$, and put

$$c^- := c_{s-1}, \qquad H := \nu_2(c - c^-), \qquad G := \max\{0, U - \nu_2(c^- x_s + m_s)\}. \tag{V4}$$

Thus $[s, u]$ is an entrance run and G is the positive part of its switch gain. We then have

$$\boxed{U \le H + G, \qquad H \le a_{s-1}, \qquad \sum_{j=s}^{u} \frac{1}{m_j} \le \kappa_c (a_{s-1} + G).} \tag{V5}$$

Consequently, divergent reciprocal mass on entrance runs forces divergence of the sum of the preceding exponents or of the positive entrance-switch gains.

For the family in (E3), the second nonpositive block satisfies

$$c = 2, \qquad B = k, \qquad U = k+2, \qquad V = 2, \qquad H = 2, \qquad G = k. \tag{V6}$$

Hence equality holds in the reciprocal-mass estimate in (V3).

Proof. Iteration of (11) through the run gives

$$2^B (c x_{u+1} + m_{u+1}) = 3^{u-s+1} (c x_s + m_s).$$

Taking 2-adic valuations proves $B = U - V$. Since every $a_j \ge 1$, the run length is at most B. If $c \in \{1, 3\}$, parity in

$$2^{a_j} m_{j+1} = 3 m_j - c$$

shows that m_j is odd. Since $m_j \ge 2$, we have $m_j \ge 3$. If $c = 2$, the same identity shows that m_j is even, so $m_j \ge 2$. Summing these pointwise bounds proves (V3).

At the entrance switch,

$$c x_s + m_s = (c^- x_s + m_s) + (c - c^-) x_s. \tag{V7}$$

Since x_s is odd, the second term on the right has valuation H. If the two terms in (V7) have distinct valuations, then U is their minimum and hence $U \le H$. If they have the same valuation, then this common valuation is H and $G = U - H$. In both cases, $U \le H + G$.

Write $d_{s-1} = 1 - c^- \geq 1$. Equation (4) gives

$$d_{s-1} \leq 2^{a_{s-1}} - 1.$$

Therefore,

$$1 \leq c - c^- = c + d_{s-1} - 1 \leq 2^{a_{s-1}} + 1 < 2^{a_{s-1}+1}.$$

It follows that $H \leq a_{s-1}$. Combining this with (V3) proves (V5). After summing (V5), a divergent left side forces at least one of the two nonnegative sums on the right to diverge.

Finally, equations (E7) and (E4) give

$$\nu_2(2x_2 + m_2) = k + 2, \qquad \nu_2(2x_{k+2} + m_{k+2}) = 2.$$

At the entrance, $c^- = -2$, so (V4) gives $H = 2$ and $G = k$. The remaining claims in (V6) follow from (E3), and (E6) gives equality in (V3). □

Corollary 11 (negative shadows of defect runs)**.** *Let $[s, u]$ be a constant-color run with*

$$c_j = c \in \{1, 2, 3\} \qquad (s \leq j \leq u). \tag{Y1}$$

The following nonempty sequences consist of positive odd integers and are accelerated $3y - 1$ paths:

c	*shadow states*	*discarded ceiling index*
1	$y_j := m_j \quad (s \leq j \leq u)$	$\varnothing$
2	$y_j := m_j/2 \quad (s \leq j < u)$	u
3	$y_j := m_j/3 \quad (s < j \leq u)$	s.

(Y2)

Thus every consecutive pair of displayed shadow states satisfies

$$\boxed{2^{a_j} y_{j+1} = 3y_j - 1, \qquad a_j = v_2(3y_j - 1).} \tag{Y3}$$

Let $H := 10.789$, the certified reciprocal-mass bound for every finite positive simple accelerated $3y - 1$ path. If the corresponding nonempty shadow in (Y2) is simple, then

$$\boxed{\sum_{j=s}^{u} \frac{1}{m_j} < \begin{cases} H, & c = 1, \\ H/2 + 1/2, & c = 2, \\ H/3 + 1/3, & c = 3. \end{cases}} \tag{Y4}$$

The same bounds hold when the indicated shadow is empty. Consequently, if a run violates its corresponding bound in (Y4), its shadow contains a repeated state and hence a positive cycle of the accelerated $3y - 1$ map.

Proof. On the run, the ceiling recurrence is

$$2^{a_j} m_{j+1} = 3m_j - c. \tag{Y5}$$

If $c = 1$, parity in (Y5) shows that every m_j in the run is odd, and (Y3) follows directly with $y_j = m_j$.

Suppose that $c = 2$. Parity in (Y5) shows that m_j is even at every index of the run. For $j < u$, put $m_j = 2y_j$. Whenever $j + 1 < u$, division of (Y5) by 2 gives

$$2^{a_j} y_{j+1} = 3y_j - 1.$$

The same identity with the next ceiling state still even shows that each displayed y_j is odd. Hence the powers 2^{a_j} are exact.

Finally, suppose that $c = 3$. Parity in (Y5) shows that every m_j in the run is odd. For $s < j \le u$, the preceding instance of (Y5) gives

$$2^{a_{j-1}} m_j = 3(m_{j-1} - 1),$$

so $3 \mid m_j$. Therefore $y_j = m_j/3$ is a positive odd integer, and division of (Y5) by 3 proves (Y3). This establishes (Y2)–(Y3).

If $c = 1$, the identity $m_j = y_j$ and the certified simple-path bound give the first line of (Y4). If $c = 2$, the shadow contributes less than $H/2$ to the ceiling reciprocal mass, while the discarded endpoint satisfies $m_u \ge 2$. If $c = 3$, the shadow contributes less than $H/3$, while m_s is an odd integer at least 3. These observations prove (Y4), including the empty-shadow cases.

If a run violates (Y4), the corresponding shadow cannot be simple. It therefore contains two equal positive odd states. The accelerated map

$$y \longmapsto \frac{3y - 1}{2^{v_2(3y-1)}}$$

is deterministic, so the segment between those equal states is a positive cycle. □

Proposition 12 (bounded aperiodic all-nonpositive countermodel)**.** *There is an aperiodic infinite strict-barrier valuation word with the following properties.*

1. *Its coefficient q_j tends to infinity.*
2. *It has one fixed real companion satisfying*

$$4 < e_j \le 14, \qquad d_j \in \{-1, 0\} \qquad (j \ge 0). \tag{C1}$$

In particular, $\sum_j 1/e_j = \infty$.

3. *Every finite prefix has a positive ordinary integer realization.*

4. *The least positive dyadic cylinder representatives of its prefixes tend to infinity. Hence the complete word is not realized by one nonnegative ordinary integer.*

Proof. We use the following three length-16 words:

$$\begin{aligned} w_U &:= 1113(12)^6, \\ w_V &:= 11221122(12)^4, \\ w_W &:= 1122121122(12)^3. \end{aligned} \tag{C2}$$

Every word in (C2) has total exponent 24. Their exponent-prefix vectors are

$$\begin{aligned} &(1,2,3,6,7,9,10,12,13,15,16,18,19,21,22,24), \\ &(1,2,4,6,7,8,10,12,13,15,16,18,19,21,22,24), \\ &(1,2,4,6,7,9,10,11,13,15,16,18,19,21,22,24). \end{aligned} \tag{C3}$$

Coordinatewise, these vectors are at most

$$(1,3,4,6,7,9,11,12,14,15,17,19,20,22,23,25) = \left(\lfloor k \log_2 3 \rfloor\right)_{k=1}^{16}. \tag{C4}$$

Thus every nonempty prefix of each word is a strict barrier. Also

$$P := 3^{16} = 43046721 > 16777216 = 2^{24} =: M. \tag{C5}$$

Each word returns the companion ceiling to $m = 5$. The complete ceiling state sequences are

$$\begin{aligned} U: &\quad (5,7,10,14,5,(7,5)^6), \\ V: &\quad (5,7,10,7,5,7,10,7,5,(7,5)^4), \\ W: &\quad (5,7,10,7,5,7,5,7,10,7,5,(7,5)^3). \end{aligned} \tag{C6}$$

The corresponding defect words are

$$\begin{aligned} \delta_U &:= (0,0,-1,-1,0^{12}), \\ \delta_V &:= (0,0,-1,0,0,0,-1,0^9), \\ \delta_W &:= (0,0,-1,0^5,-1,0^7). \end{aligned} \tag{C7}$$

Indeed, direct substitution in

$$2^{a_j} m_{j+1} = 3m_j - 1 + d_j$$

verifies (C6) and (C7).

Let θ be the ceiling error at a block boundary. Iterating $2^{a_j}\theta_{j+1} = 3\theta_j + d_j$ gives a forward affine map on each block. The admissible starting interval, the resulting image interval, and the inverse map are as follows:

	domain	image	$\phi_i(t)$
U	$[20/81, 4/9)$	$[0, 531441/1048576)$	$(Mt + 10628820)/P$
V	$[580/2187, 4/9)$	$[0, 964467/2097152)$	$(Mt + 11416140)/P$
W	$[4964/19683, 4/9)$	$[0, 1034451/2097152)$	$(Mt + 10856268)/P.$

(C8)

For completeness, these intervals follow from the prefix identity

$$2^{B_i}\theta_i = 3^i\theta_0 - R_i. \tag{C9}$$

At every prefix we impose $0 \le \theta_i < 1$, take the maximum of the lower bounds $R_i/3^i$, and take the minimum of the upper bounds $(R_i + 2^{B_i})/3^i$. This gives (C8) by rational arithmetic.

Put

$$\begin{aligned} p_U &:= \frac{2125764}{5253901}, \\ p_V &:= \frac{78732}{181169}, \qquad I := [p_U, p_V]. \\ p_W &:= \frac{638604}{1545265}, \end{aligned} \tag{C10}$$

These are the respective fixed points of the three inverse maps in (C8), and

$$p_U < p_W < p_V.$$

Direct cross multiplication in (C8) shows that I lies in all three domains and all three image intervals. Each ϕ_i is increasing, has contraction ratio $M/P < 1$, and has its fixed point in I. Therefore

$$\phi_i(I) \subseteq I \qquad (i \in \{U, V, W\}). \tag{C11}$$

Let $\omega = (\omega_0, \omega_1, \ldots)$ be any infinite word over $\{U, V, W\}$. For every k, the limit

$$\theta_k := \lim_{L\to\infty} \phi_{\omega_k} \circ \phi_{\omega_{k+1}} \circ \cdots \circ \phi_{\omega_L}(p_U) \tag{C12}$$

exists in I, since all three maps have the same contraction ratio less than 1. It satisfies

$$\theta_k = \phi_{\omega_k}(\theta_{k+1}).$$

Thus concatenating the blocks selected by ω gives one fixed real companion with the ceiling states and defects in (C6) and (C7). This proves (C1). Every block is a strict barrier and multiplies the coefficient by $P/M > 1$, so the complete concatenation is a strict barrier and $q_j \to \infty$.

It remains to choose the driver so that it is aperiodic and has no ordinary integer seed. Enumerate all eventually periodic ternary words as $\eta^{(0)}, \eta^{(1)}, \ldots$. Suppose a finite driver has been chosen, with valuation prefix v and least positive cylinder representative $n(v)$. The three extensions vw_U, vw_V, vw_W have the same length and total exponent, and their exact cylinders are pairwise disjoint. Hence $n(v)$ realizes at most one of them. At least two choices have a new least representative different from $n(v)$. At stage k, choose one of those two symbols which is also different from the k-th symbol of $\eta^{(k)}$.

The resulting driver differs from every eventually periodic ternary word, so it is aperiodic. Since the three codewords are distinct and have the same length, eventual periodicity of their concatenation would force the aligned codeword sequence to be eventually periodic. Hence the valuation word is also aperiodic. Cylinder compatibility gives

$$n(vw_i) = n(v) + h\,2^{A(v)+1} \qquad (h \geq 1) \tag{C13}$$

at every stage. Therefore the least representatives tend to infinity. Every finite prefix is nevertheless realized by its own least positive representative. Finally, $e_j \leq 14$ gives $\sum_j 1/e_j = \infty$. □

Proposition 13 (arbitrarily long zero-defect finite windows)**.** *For every $k \geq 1$, the strict-barrier word*

$$w_k := (1,2)^k \tag{23}$$

is realized by a positive integer orbit segment and by a synchronized real companion for which

$$e_{2i} = 5, \qquad e_{2i+1} = 7, \qquad d_j = 0 \quad (0 \leq j < 2k). \tag{24}$$

Nevertheless, if $D(w) := C(w) + 2^{A(w)}$, then

$$v_3(D(w_k)) = 0. \tag{25}$$

Thus the terminal 3-adic deficit of this block is its full length $2k$.

Proof. For $0 \le i \le k$, define

$$x_{2i} := 2 \cdot 9^i 8^{k-i} - 5, \tag{26}$$

and, for $0 \le i < k$, define

$$x_{2i+1} := 3 \cdot 9^i 8^{k-i} - 7. \tag{27}$$

Every displayed integer is positive. If $N_i := 2 \cdot 9^i 8^{k-i}$, then

$$3x_{2i} + 1 = 3N_i - 14 \equiv 2 \pmod 4,$$

so the valuation at x_{2i} is exactly 1. Moreover

$$3x_{2i+1} + 1 = 9^{i+1} 8^{k-i} - 20 \equiv 4 \pmod 8,$$

so the valuation at x_{2i+1} is exactly 2. Division gives (26) and (27) at the following states. Hence the segment realizes w_k exactly.

The companion values in (24) satisfy

$$\frac{3 \cdot 5 - 1}{2} = 7, \qquad \frac{3 \cdot 7 - 1}{4} = 5.$$

They are integers, so $m_j = e_j$, $\theta_j = 0$, and (3) gives $d_j = 0$. Adding the two paired recurrences also gives

$$x_j + e_j = q_j(x_0 + 5)$$

throughout the segment.

For every even prefix of length $2i$ we have $2^A = 8^i < 9^i = 3^{2i}$. For every odd prefix of length $2i + 1$ we have

$$2^A = 2 \cdot 8^i < 3 \cdot 9^i = 3^{2i+1}.$$

Thus every prefix is a strict barrier.

For the block $(1, 2)$, the coefficient pair and carry are

$$P = 9, \qquad M = 8, \qquad C = 5.$$

The concatenation formula therefore gives

$$C(w_k) = 5\frac{9^k - 8^k}{9 - 8} = 5(9^k - 8^k).$$

It follows that

$$D(w_k) = 5 \cdot 9^k - 4 \cdot 8^k \equiv -4(-1)^k \not\equiv 0 \pmod 3.$$

This proves (25). □

Corollary 14 (no finite-window bridge to deficit two or three)**.** *For every $L \geq 1$, there is a paired strict-barrier block of length at least L whose rounding defects are all nonpositive, but whose terminal 3-adic deficit is greater than 3. Therefore the hypotheses of a valid bridge from nonpositive rounding-defect mass to a deficit-two or deficit-three terminal congruence must use one fixed ordinary orbit at unboundedly many depths. No implication from the corresponding properties of one finite block can supply that bridge.*

Proof. Choose $k \geq 2$ with $2k \geq L$ and apply the proposition. Its terminal deficit is $2k \geq 4$. □